MODEL FAIRNESS PASSPORT

Brain-age model SFCN · IXI

An independent audit of accuracy, group fairness, acquisition robustness, explanations and uncertainty.

Passport	BAF-XAI-2026-001
Issued	2026-10-17
Checkpoint	sha256 13ddd261…
Valid for	IXI-like T1 MRI

ARCHITECTURE	PRETRAINING	FINE-TUNED ON	EVALUATED ON
SFCN, 3D CNN	**UK Biobank**	**IXI, 339 adults**	**73 test + 73 val**
72 age bins (18–90)	Peng et al. 2021	3 London sites	Test set used once

VERDICT

4 of 6 checks passed

● PASS	● PASS	● FAIL	● FAIL	● PASS	● PASS
Accuracy	Fairness by sex	Fairness by site	Scanner invariance	Faithful explanations	Honest uncertainty

PERFORMANCE & FAIRNESS

Check	Result	95% CI
● Test MAE · r 0.966	3.71 y	3.10 – 4.38
● Gap, male vs female	−0.59 y	−2.67 – +1.64
● Gap, HH vs Guys	+1.06 y	−1.44 – +3.53
● **Gap, IOP vs Guys**	−5.06 y	−8.66 – −1.34
● Error, any group	no diff.	all CIs span 0
● **IOP unseen, extra error**	+3.30 y	+1.62 – +4.93
● HH unseen, extra error	+0.25 y	−0.88 – +1.36

ROBUSTNESS, EXPLANATIONS, UNCERTAINTY

Check	Result
● **Site read from features**	91% chance 33%
● **Scan look alone shifts age**	3–4 y 100% same dir.
● IOP offset explained by intensity / blur / fingerprint	No ≤1 / 0 / persists
● Sanity check, randomised layers	ρ 0.13 maps destroyed
● Deletion, top vs random 5%	4.83 vs 1.04 y
● Uncertainty on unseen IOP	Doubles 92% coverage

● Pass ● Fail ● Unresolved

SUITABLE FOR

Research on brain ageing within a single acquisition site

Comparing groups scanned on the same scanner and protocol

Teaching and methods development

NOT SUITABLE FOR

Any clinical or individual decision

Comparing brain-age gaps *across* scanners without site-aware validation

New sites, without checking their uncertainty first

KNOWN LIMITATIONS

01

Small subgroups (9 IOP test subjects); single seed and split. The IOP offset still replicates in validation (−4.1) and test (−5.1).

02

In IXI, site is confounded with who each site recruited: the cause of the IOP offset (untested scanner properties or cohort) cannot be separated.

03

Healthy volunteers aged 20–86 only. Attribution resolution 8 mm. Uncertainty is the model's own spread, not a calibrated interval.

Evidence: every figure on this page comes from the scripts in scripts/03b–14 and is logged with dates in PROJECT_GUIDE.md. Statistics: age-adjusted OLS with HC3 errors, 5,000-sample bootstrap CIs, FDR where several regions are tested. Wording: acquisition-site effects, not "hospital bias"; no clinical claims.